

ELITE MATHEMATICS

Pearson Edexcel International A Level

Pure 1

Adaptive Classified

Chapter 05 · Polynomials

With Answers

Questions with full worked solutions

- 20 pages
- Past papers up to June 2026

Polynomials

Your classified route

6

questions

51

marks

3

families

7

variants

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CLASSIFIED BY IDEA

Move from recognition to method, then use the working space to complete the solution.

Polynomials

Families and variants

F01 • Factor and Solve Cubics

V01 Determine a Parameter, then Factor the Cubic

V02 Common Factor, then Quadratic Factor

F02 • Build a Polynomial from Roots and Graph Behaviour

V01 Distinct Roots and a Point

V02 Repeated Root from Touch Behaviour

V03 Intersection by Preserving Common Factors

F03 • Transfer Roots Through Transformations or Substitutions

V01 Input-transformation Root Map

V02 Substituted-variable Root Transfer with a Domain Filter

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Factor and Solve Cubics

WHAT THIS FAMILY ASKS

Determine any required coefficient, extract a common factor and factor the remaining quadratic to obtain every real root of a polynomial of degree at most three.

VARIANT ROUTE

V01 Determine a Parameter, then Factor the Cubic

V02 Common Factor, then Quadratic Factor

2 Complete Classified questions

16 total marks

2.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

$$f(x) = ax^3 + (6a + 8)x^2 - a^2x$$

where a is a positive constant.

Given $f(-1) = 32$

(a) (i) show that the only possible value for a is 3

(ii) Using $a = 3$ solve the equation

$$f(x) = 0 \quad (5)$$

(b) Hence find all real solutions of

(i) $3y + 26y^{\frac{2}{3}} - 9y^{\frac{1}{3}} = 0$

(ii) $3(9^{3z}) + 26(9^{2z}) - 9(9^z) = 0$ (5)

(Total 10 marks)

Question 1

WMA11/01 May/June 2021

T05 Polynomials

F01 Factor and Solve Cubics

V01 Determine a Parameter,
then Factor the Cubic

1

Use $f(-1) = 32$ $f(-1) = -a + (6a + 8) + a^2 = a^2 + 5a + 8$. Since $f(-1) = 32$, $a^2 + 5a - 24 = 0$.

2

Choose the positive value of a $a^2 + 5a - 24 = (a - 3)(a + 8) = 0$. Since a is positive, the only possible value is $a = 3$.

3

Solve $f(x) = 0$ Using $a = 3$, $f(x) = 3x^3 + 26x^2 - 9x = x(3x^2 + 26x - 9) = x(3x - 1)(x + 9)$.

4

List the roots $x = 0$, $x = \frac{1}{3}$, or $x = -9$.

5

Solve the equation in y Let $u = y^{1/3}$. Then $y = u^3$ and $y^{2/3} = u^2$, so $3y + 26y^{2/3} - 9y^{1/3} = 0$ becomes $3u^3 + 26u^2 - 9u = 0$.

6

Convert back to y From part (a), $u = -9, 0, \frac{1}{3}$. Therefore $y = u^3$, giving $y = -729, 0, \frac{1}{27}$.

7

Solve the exponential equationLet $u = 9^z$. Then $3(9^{3z}) + 26(9^{2z}) - 9(9^z) = 0$ becomes $3u^3 + 26u^2 - 9u = 0$.

8

Use positivity of 9^z

Since $9^z > 0$, the only valid root is $u = \frac{1}{3}$. Hence $9^z = \frac{1}{3}$, so $(3^2)^z = 3^{-1}$ and $2z = -1$. Therefore $z = -\frac{1}{2}$.

FINAL ANSWER

(a)(i) $a = 3$. (a)(ii) $x = -9, 0, \frac{1}{3}$. (b)(i) $y = -729, 0, \frac{1}{27}$. (b)(ii) $z = -\frac{1}{2}$.

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4.

In this question you must show detailed reasoning.
Solutions relying on calculator technology are not acceptable.

$$f(x) = x^2(2x + 1) - 15x$$

(a) Solve

$$f(x) = 0 \quad (4)$$

(b) Hence solve

$$y^{\frac{4}{3}}\left(2y^{\frac{2}{3}} + 1\right) - 15y^{\frac{2}{3}} = 0 \quad y > 0$$

giving your answer in simplified surd form. (2)

(Total for Question 4 is 6 marks)

Question 2

WMA11/01 October 2022

T05 Polynomials

F01 Factor and Solve Cubics

V02 Common Factor, then
Quadratic Factor

1

Factor $f(x)$

$$f(x) = x^2(2x + 1) - 15x = x(2x^2 + x - 15).$$

2

Factor the quadratic

$$2x^2 + x - 15 = (2x - 5)(x + 3).$$

3

Solve $f(x) = 0$

$$x(2x - 5)(x + 3) = 0, \text{ so } x = 0, x = \frac{5}{2}, \text{ or } x = -3.$$

4

Use the substitution for part (b)

The equation is the same as $f(u) = 0$ with $u = y^{2/3}$.

5

Choose the valid root

Since $y > 0$, $u = y^{2/3} > 0$. The valid root is $u = \frac{5}{2}$, so $y^{2/3} = \frac{5}{2}$.

6

Write y in surd form

$$y = \left(\frac{5}{2}\right)^{3/2} = \sqrt{\frac{125}{8}} = \frac{5\sqrt{10}}{4}.$$

FINAL ANSWER

$$(a) x = -3, 0, \frac{5}{2}. \quad (b) y = \frac{5\sqrt{10}}{4}.$$

Build a Polynomial from Roots and Graph Behaviour

WHAT THIS FAMILY ASKS

Translate x-intercepts and touch/cross behaviour into factors with the correct multiplicities, determine a scale from another point, and exploit factor structure in later intersections.

VARIANT ROUTE

V01 Distinct Roots and a Point

V02 Repeated Root from Touch Behaviour

V03 Intersection by Preserving Common Factors

4 Complete Classified questions

35 total marks

8. The curve C_1 has equation

$$y = 3x^2 + 6x + 9$$

(a) Write $3x^2 + 6x + 9$ in the form

$$a(x + b)^2 + c$$

where a , b and c are constants to be found.

(3)

The point P is the minimum point of C_1

(b) Deduce the coordinates of P .

(1)

A different curve C_2 has equation

$$y = Ax^3 + Bx^2 + Cx + D$$

where A , B , C and D are constants.

Given that C_2

- passes through P
- intersects the x -axis at -4 , -2 and 3

(c) find, making your method clear, the values of A , B , C and D .

(5)

(Total 9 marks)

Question 3

WMA11/01 May/June 2021

T05 Polynomials

F02 Build a Polynomial from
Roots and Graph BehaviourV01 Distinct Roots and a
Point

1

Complete the square

$$3x^2 + 6x + 9 = 3(x^2 + 2x) + 9 = 3((x + 1)^2 - 1) + 9 = 3(x + 1)^2 + 6.$$

2

Deduce the minimum point

The minimum occurs when $(x + 1)^2 = 0$, so $x = -1$ and $y = 6$. Therefore $P = (-1, 6)$.

3

Use the x-intercepts of C2

Since C_2 intersects the x-axis at $-4, -2$ and 3 , it has form $y = k(x + 4)(x + 2)(x - 3)$.

4

Use point P to find k

Substitute $P = (-1, 6)$: $6 = k(3)(1)(-4) = -12k$, so $k = -\frac{1}{2}$.

5

Expand the cubic

$$(x + 4)(x + 2)(x - 3) = (x^2 + 6x + 8)(x - 3) = x^3 + 3x^2 - 10x - 24.$$

6

Read off A, B, C and D

Thus $y = -\frac{1}{2}x^3 - \frac{3}{2}x^2 + 5x + 12$, so $A = -\frac{1}{2}$, $B = -\frac{3}{2}$, $C = 5$ and $D = 12$.

FINAL ANSWER

(a) $3(x + 1)^2 + 6$. (b) $P = (-1, 6)$. (c) $A = -\frac{1}{2}$, $B = -\frac{3}{2}$, $C = 5$, $D = 12$.

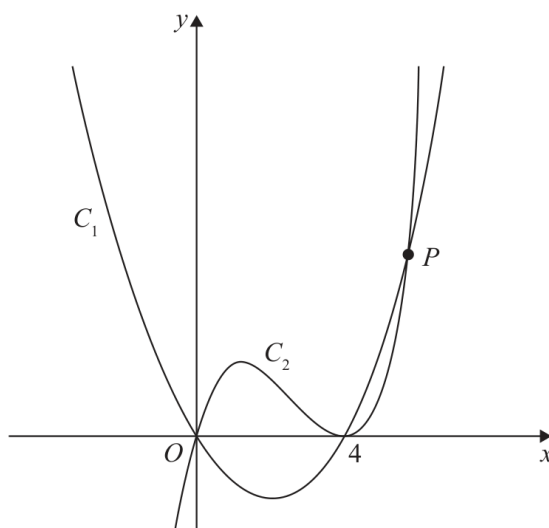


Figure 1

Figure 1 shows a sketch of part of the curves C_1 and C_2

Given that C_1

- has equation $y = f(x)$ where $f(x)$ is a quadratic function
- cuts the x -axis at the origin and at $x = 4$
- has a minimum turning point at $(2, -4.8)$

(a) find $f(x)$

(3)

Given that C_2

- has equation $y = g(x)$ where $g(x)$ is a cubic function
- cuts the x -axis at the origin and meets the x -axis at $x = 4$
- passes through the point $(6, 7.2)$

(b) find $g(x)$

(3)

The curves C_1 and C_2 meet in the first quadrant at the point P , shown in Figure 1.

(c) Use algebra to find the coordinates of P .

(4)

(Total for Question 4 is 10 marks)

Question 4

WMA11/01 October 2024

T05 Polynomials

F02 Build a Polynomial from
Roots and Graph BehaviourV02 Repeated Root from
Touch Behaviour

1

Use the roots of C1Since C_1 cuts the x-axis at 0 and 4, let $f(x) = ax(x - 4)$.

2

Use the minimum point of C1At $x = 2$, $f(2) = -4a = -4.8 = -\frac{24}{5}$, so $a = \frac{6}{5}$.

3

Write f(x)

$$f(x) = \frac{6}{5}x(x - 4).$$

4

Use the roots of C2The cubic cuts the x-axis at the origin and meets the x-axis at $x = 4$, so $x = 4$ is a repeated root. Let $g(x) = bx(x - 4)^2$.

5

Use the point (6, 7.2)

$$7.2 = b(6)(2^2) = 24b, \text{ so } b = 0.3 = \frac{3}{10}.$$

6

Equate the curves

$$\frac{6}{5}x(x - 4) = \frac{3}{10}x(x - 4)^2.$$

7

Find the first-quadrant intersection

Ignoring the shared roots on the x-axis, divide by $x(x - 4)$ to get $\frac{6}{5} = \frac{3}{10}(x - 4)$.
Hence $x = 8$.

8

Find y

$$y = f(8) = \frac{6}{5}(8)(4) = \frac{192}{5}.$$

FINAL ANSWER

$$(a) f(x) = \frac{6}{5}x(x - 4). \quad (b) g(x) = \frac{3}{10}x(x - 4)^2. \quad (c) P = (8, \frac{192}{5}).$$

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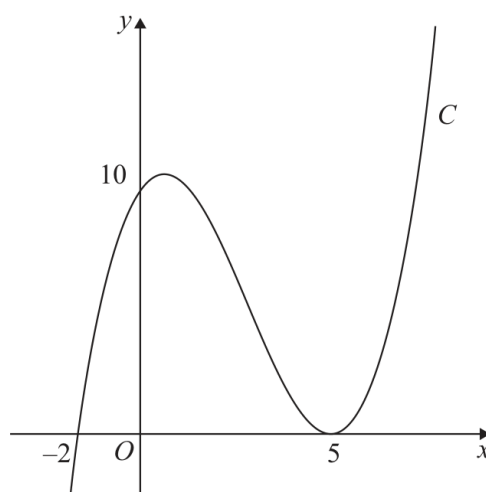


Figure 3

Figure 3 shows a sketch of the curve C with equation $y = f(x)$ where $f(x)$ is a cubic function in x .

The curve C

- cuts the x -axis at $(-2, 0)$ and cuts the y -axis at $(0, 10)$
- touches the x -axis at $(5, 0)$

as shown in Figure 3.

(a) Deduce the roots of the equation

(i) $f\left(\frac{1}{3}x\right) = 0$

(ii) $f(x - 3) = 0$

(2)

(b) Find an expression for $f(x)$. You should leave your answer in factorised form.

(3)

The curve C intersects the straight line $y = 10(x + 2)$ at exactly three points.

(c) Use algebra to find the exact x coordinates of the three points of intersection.

(Solutions based entirely on calculator technology are not acceptable.)

(4)

(Total for Question 6 is 9 marks)

Question 5

WMA11/01 January 2026

T05 Polynomials

F02 Build a Polynomial from
Roots and Graph BehaviourV03 Intersection by Preserv-
ing Common Factors

1

Roots of f one third x

The roots of $f(x) = 0$ are $x = -2$ and $x = 5$, with $x = 5$ repeated. For $f\left(\frac{1}{3}x\right) = 0$, set $\frac{1}{3}x = -2$ or 5 , giving $x = -6$ or 15 .

2

Roots of f of x minus 3

For $f(x - 3) = 0$, set $x - 3 = -2$ or 5 , giving $x = 1$ or 8 .

3

Use the roots and repeated root

Since the curve cuts at $(-2, 0)$ and touches at $(5, 0)$, write $f(x) = k(x + 2)(x - 5)^2$.

4

Use the y-intercept

The curve passes through $(0, 10)$, so $10 = k(2)(25) = 50k$.

5

State $f(x)$

$k = \frac{1}{5}$, hence $f(x) = \frac{1}{5}(x + 2)(x - 5)^2$.

6

Set equal to the line

$\frac{1}{5}(x + 2)(x - 5)^2 = 10(x + 2)$.

7

First solution

The factor $x + 2$ gives one solution $x = -2$.

8

Find the other solutions

For $x \neq -2$, divide by $x + 2$: $\frac{1}{5}(x - 5)^2 = 10$, so $(x - 5)^2 = 50$.

9

State all intersections

$x = 5 \pm \sqrt{50} = 5 \pm 5\sqrt{2}$. Therefore the three x-coordinates are $-2, 5 - 5\sqrt{2}$ and $5 + 5\sqrt{2}$.

FINAL ANSWER

(a)(i) $-6, 15$. (a)(ii) $1, 8$. (b) $f(x) = \frac{1}{5}(x + 2)(x - 5)^2$. (c) $x = -2, 5 \pm 5\sqrt{2}$.

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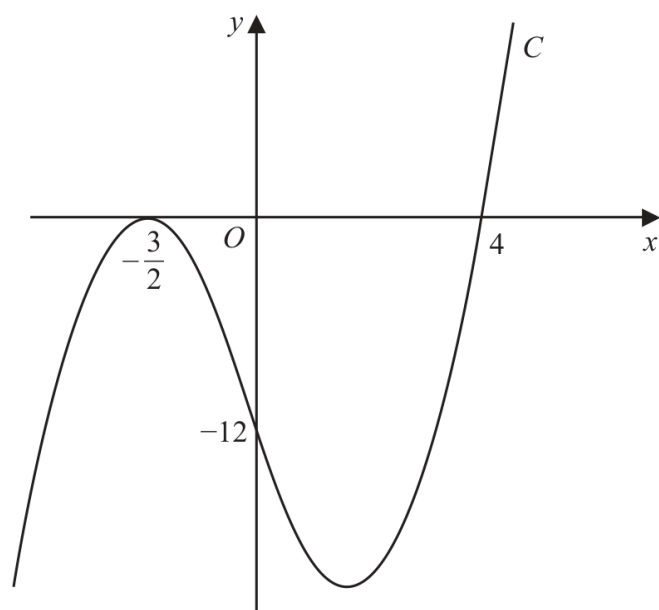


Figure 2

Figure 2 shows a sketch of the curve C with equation $y = f(x)$.

The curve C

- touches the x -axis at $-\frac{3}{2}$
- cuts the x -axis at 4
- cuts the y -axis at -12

Given that $f(x)$ is a cubic function,

(a) find $f(x)$. You may leave the answer in a factorised form.

(3)

The line with equation $y = 2x - 12$ cuts C at $(0, -12)$ and at two further points, P and Q .

(b) Use algebra to find the exact x coordinates of P and Q .
Give each answer in simplest form.

(Solutions relying entirely on calculator technology are not acceptable.)

(4)

(Total for Question 6 is 7 marks)

Question 6

WMA11/01 May/June 2026

T05 Polynomials

F02 Build a Polynomial from
Roots and Graph BehaviourV03 Intersection by Preserv-
ing Common Factors

1

(a) Build the cubic from its roots

The curve touches the x-axis at $x = -\frac{3}{2}$, so this is a repeated root. It crosses the x-axis at $x = 4$, so this is a simple root. Hence

$$f(x) = k(2x + 3)^2(x - 4).$$

2

(a) Use the y-intercept

The curve passes through $(0, -12)$. Substituting $x = 0$ gives

$$-12 = k(3)^2(-4) = -36k,$$

so $k = \frac{1}{3}$. Therefore

$$f(x) = \frac{1}{3}(2x + 3)^2(x - 4).$$

3

(b) Form the intersection equation

At an intersection with the line $y = 2x - 12$,

$$\frac{1}{3}(2x + 3)^2(x - 4) = 2x - 12.$$

Multiplying by 3 and expanding gives

$$(4x^2 + 12x + 9)(x - 4) = 6x - 36,$$

so

$$4x^3 - 4x^2 - 45x = 0.$$

4

(b) Remove the known intersection and solveFactor out the known root $x = 0$:

$$x(4x^2 - 4x - 45) = 0.$$

The two further intersections satisfy

$$4x^2 - 4x - 45 = 0.$$

Using the quadratic formula,

$$x = \frac{4 \pm \sqrt{(-4)^2 - 4(4)(-45)}}{8} = \frac{4 \pm 4\sqrt{46}}{8} = \frac{1 \pm \sqrt{46}}{2}.$$

FINAL ANSWER

$$(a) f(x) = \frac{1}{3}(2x + 3)^2(x - 4). \quad (b) x = \frac{1 - \sqrt{46}}{2} \text{ or } x = \frac{1 + \sqrt{46}}{2}.$$